QRT812 \ QRT812F \ QRT812D

PLANAR STRUCTURED SUPERFAST RECOVERY RECTIFIERS

Voltage	1200 V	Current	8 A

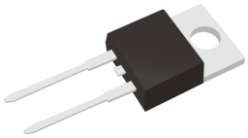
Features

- Planar structure with EPI wafer
- Hyperfast recovery time, reduced Qrr and soft recovery
- For PFC CCM operation
- Low leakage current
- Plastic package has Underwriters Laboratory Flammability Classification 94V-O
  Flame Retardant Epoxy Molding Compound
- Exceeds environmental standards of MIL-S-19500/228
- Lead free in compliance with EU RoHS 2011/65/EU directive

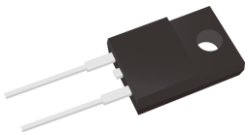
Mechanical Data

- Case: TO-220AC, ITO-220AC, TO-263 package
- Terminals: solder plated, solderable per MIL-STD-750,Method 2026
- TO-220AC Weight: 0.067 ounces, 1.89 grams
- ITO-220AC Weight: 0.055 ounces, 1.56 grams.
- TO-263 Weight: 0.051 ounces, 1.46 grams.
- Marking: Part number

QRT812 TO-220AC



QRT812F ITO-220AC



QRT812D TO-263



Maximum Ratings And Electrical Characteristics (T_A=25°C unless otherwise noted)

PARAMETER		SYMBOL	VALUE	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	1200	V
Maximum rms voltage		V_{RMS}	840	V
Maximum dc blocking voltage		V_R	1200	V
Maximum average forward rectified current		$I_{F(AV)}$	8	A
Peak forward surge current : 8.3ms single half sine-wave superimposed on rated load		I_{FSM}	90	A
Typical thermal resistance	TO-220AC (Note 1)	$R_{\theta JC}$	2	°C/W
	ITO-220AC (Note 1)		5.5	
	TO-263 (Note 1)		2	
Operating junction temperature range		T_J	-55 to +175	°C
Storage temperature range		T_{STG}	-55 to +175	°C

Note : 1. Device mounted on a infinite heatsink , then measured the center of the marking side.

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Electrical Characteristics (T_A=25°C unless otherwise noted)

PARAMETER	SYMBOL	TEST CONDITION		MIN.	TYP.	MAX.	UNITS
Breakdown voltage	V_{BR}	I_R=100μA	T_J=25°C	1200	-	-	V
Instantaneous forward voltage	V_F	I_F=1A	T_J=25°C	-	1.46	-	V
		I_F=5A		-	2.06	-	
		I_F=8A		-	2.29	3.2	
		I_F=1A	T_J=125°C	-	0.99	-	V
		I_F=5A		-	1.58	-	
		I_F=8A		-	1.82	-	
Reverse current	I_R	V_R=1200V	T_J=25°C	-	-	3	μA
			T_J=125°C	-	5	-	μA
Reverse recovery time	T_{RR}	I_F=1A V_R=30V di/dt=100A/μs	T_J=25°C	-	-	40	ns
		I_F=8A V_R=400V di/dt=200A/μs	T_J=25°C	-	45	-	ns
Peak recovery current	I_{RRM}	I_F=8A V_R=400V di/dt=200A/μs	T_J=25°C	-	3.9	-	A
Reverse recovery charge	Q_{RR}	I_F=8A V_R=400V di/dt=200A/μs	T_J=25°C	-	87.7	-	nC

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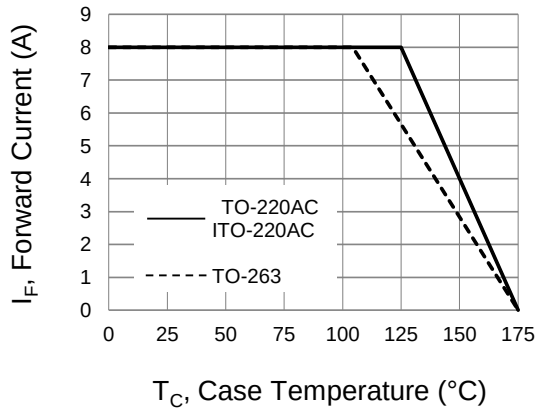


Fig.1 Forward Current Derating Curve

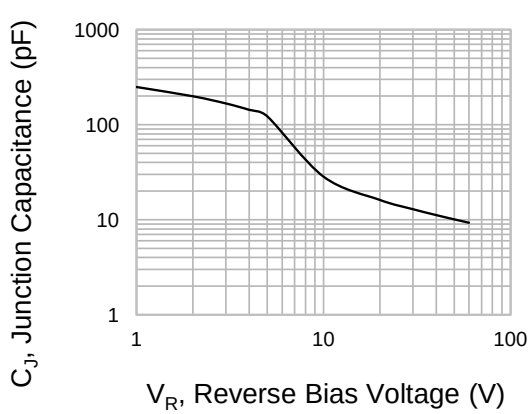


Fig.2 Typical Junction Capacitance

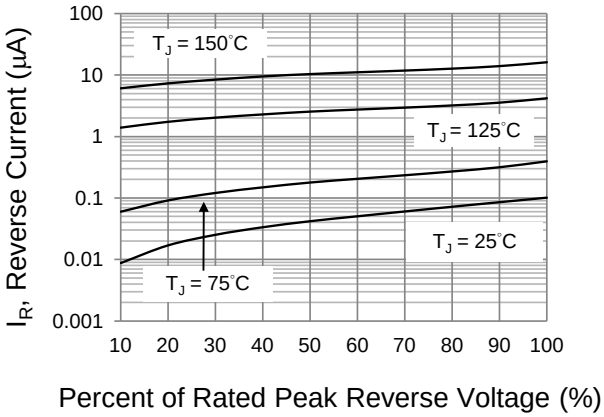


Fig.3 Typical Reverse Characteristics

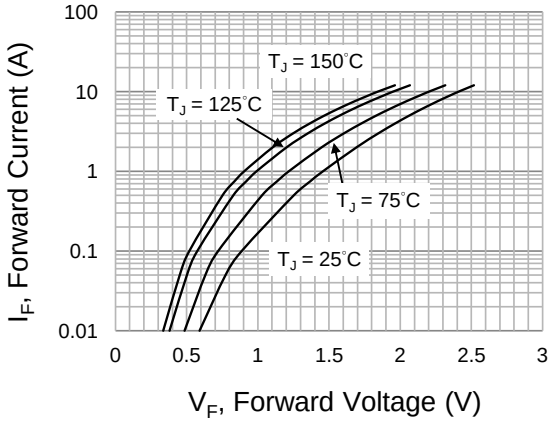


Fig.4 Typical Forward Characteristics

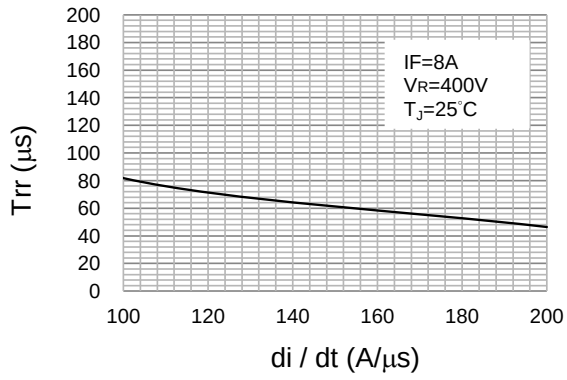


Fig.5 Typical Reverse Recovery Time Versus di/dt

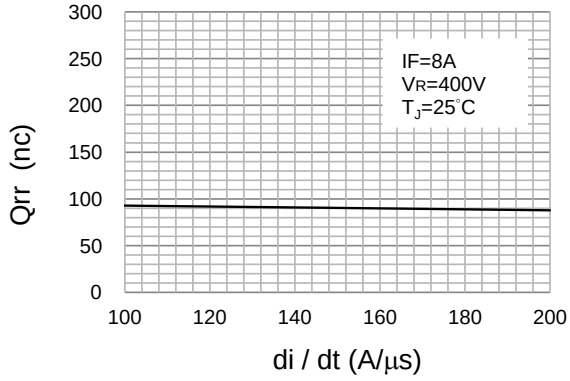
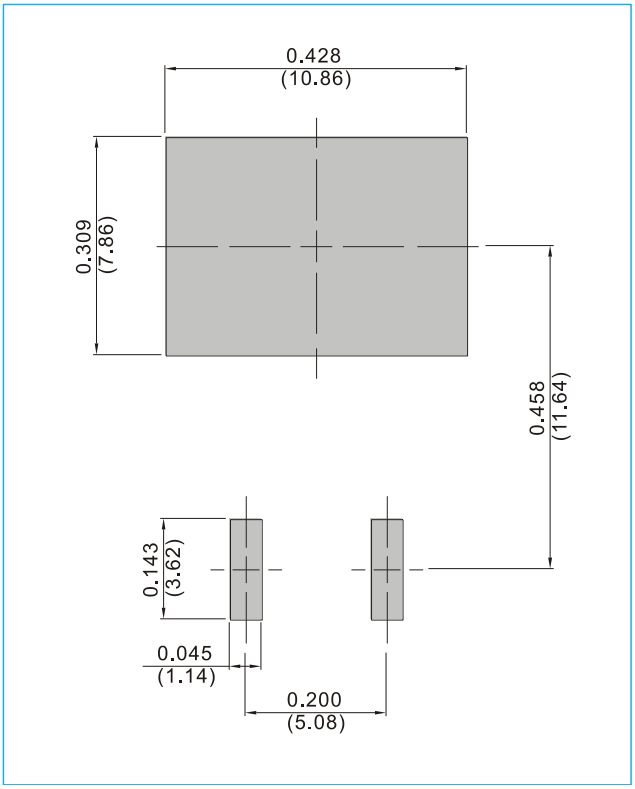


Fig.6 Typical Reverse Recovery Charges Versus di/dt

QRT812 \ QRT812F \ QRT812D

MOUNTING PAD LAYOUT

TO-263 / D²PAK Unit：inch(mm)



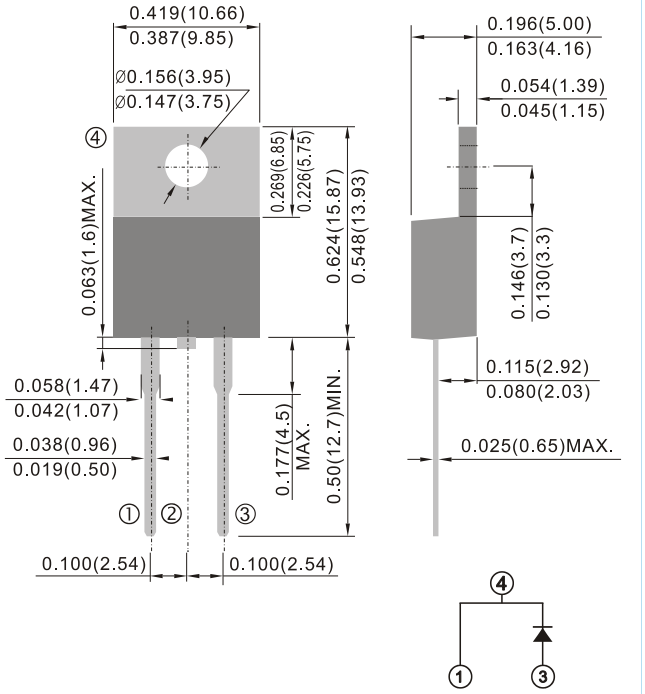
ORDER INFORMATION

- Packing information
  T/R – 0.8K per 13" plastic Reel

QRT812 \ QRT812F \ QRT812D

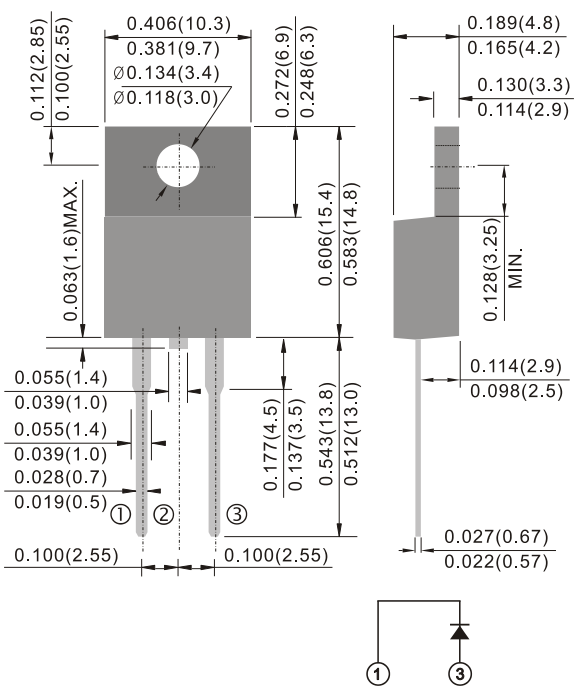
TO-220AC

Unit : inch(mm)



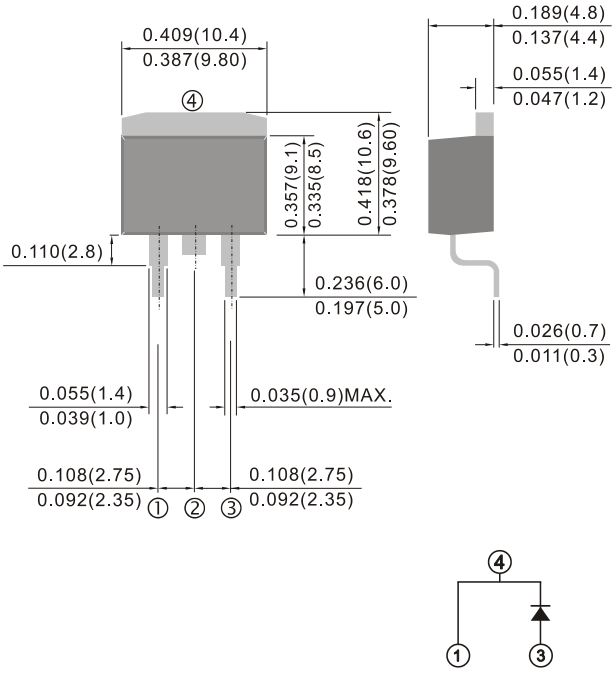
ITO-220AC

Unit : inch(mm)



TO-263 / D²PAK

Unit : inch(mm)



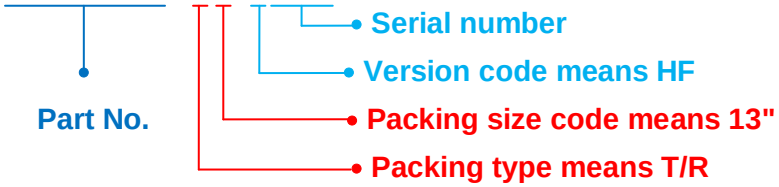
QRT812 \ QRT812F \ QRT812D

Part No_packing code_Version

QRT812_T0_00001
QRT812_T0_10001
QRT812F_T0_00001
QRT812F_T0_10001
QRT812D_R2_00001
QRT812D_R2_10001

For example :

RB500V-40_R2_00001



Packing Code XX				Version Code XXXXX		
Packing type	1st Code	Packing size code	2nd Code	HF or RoHS	1st Code	2nd~5th Code
Tape and Ammunition Box (T/B)	A	N/A	0	HF	0	serial number
Tape and Reel (T/R)	R	7"	1	RoHS	1	serial number
Bulk Packing (B/P)	B	13"	2			
Tube Packing (T/P)	T	26mm	X			
Tape and Reel (Right Oriented) (TRR)	S	52mm	Y			
Tape and Reel (Left Oriented) (TRL)	L	PANASERT T/B CATHODE UP (PBCU)	U			
FORMING	F	PANASERT T/B CATHODE DOWN (PBCD)	D			

QRT812 \ QRT812F \ QRT812D

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